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① $P(E) = 1/4$, $P(F) = 1/8$, $P(G) = 5/8$, and $P(H) = 3/8$. The events E and F are mutually disjoint whereas the events G and H are statistically independent.

ⓐ Are events E and F statistically independent

E and F are said to be mutually disjoint $\therefore E \cap F = 0$

For a statistically independent event $P(E \cap F) = P(E) \times P(F)$.

Earlier stated that E and F are mutually disjoint $P(E \cap F) = 0$

for this to support E and F being statistically independent either $P(E)$ or $P(F) = 0$

from the question $P(E) = 1/4$, $P(F) = 1/8$ [non-zero probability].

$\therefore P(E \cap F) = 0 \neq P(E) \times P(F)$. i.e if 2 events are mutually disjoint and both have a non-zero probability, they cannot be statistically independent.

ⓑ Determine $P(E \cup F)$ and $P(E \cap F)$.

$$P(E \cup F) = P(E) + P(B) - P(E \cap F).$$
$$= 1/4 + \frac{1}{8} - 0$$
$$= \frac{2+1}{8}$$
$$= 3/8$$

$$P(E \cap F) = 0$$

(c) Are the events G^c and H^c statistically independent ?

Stating De Morgan's law $A' \cap B' = (A \cup B)'$

$P(G^c \cap H^c) = P((G \cup H)^c)$ ⟶ For this to be true $P(G^c \cap H^c) = P(G)^c \cdot P(H)^c$

$P((G \cup H)^c) = \cancel{P(1 - (G \cup H))} \cdot 1 - P(G \cup H)$ --- (i)

$P(G \cup H) = P(G) + P(H) - P(G \cap H)$.

Since G & H are statistically independent.

$P(G \cup H) = P(G) + P(H) - P(G) \cdot P(H)$. — substituting in eqn 1

$P((G \cup H)^c) = 1 - [P(G) + P(H) - P(G) \cdot P(H)]$

$= 1 - P(G) - P(H) + P(G) \cdot P(H)$.

$= (1 - P(G)) \times (1 - P(H))$.

$(1 - P(G)) = \cancel{P(G)^c} \; P(G^c)$.

$(1 - P(H)) = P(H^c)$.

$\therefore P((G \cup H)^c) = P(G^c \cap H^c) = P(G^c) \cdot P(H^c)$. Showing that bothe events are statistically independent.

(d) Determine $P(E|F)$, the probability of event E conditioned on Event F.

$$P(E|F) = \frac{P(E \cap F)}{P(F)}.$$

Earlier stated event E and F are mutually disjoint.

$\therefore P(E \cap f) = 0$

$$P(E|F) = \frac{0}{P(F)}$$

$$= 0$$

This is to say, for mutually disjoint events E and F $P(E|F) = 0$. and $P(F|E) = 0$.

(2) X denotes a random Variable with uniform probability density function over the interval [-8, +7]. Let A denote the event $\{|X| \leq 4\}$.

(a) Conditional Expected Value $E[X|A]$

(b) Conditional Variance $Var[X|A]$

Soln

N.B: Interval = [-8, +7]

Recalling PDF for Uniform Distribution $f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x < b \\ 0 & \text{otherwise.} \end{cases}$

$$f_X(x) = \begin{cases} \frac{1}{7-(-8)} & -8 \leq x < 7 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{15} & -8 \leq x < 7 \\ 0 & \text{otherwise.} \end{cases}$$

A denotes the event $\{|X| \leq 4\}$ = $-4 \leq x \leq 4$.

$$P(A) = P(x \leq 4) - P(x \leq -4)$$

$$= \frac{13}{15} - \frac{5}{15}$$

$$= \frac{8}{15} //$$

(a) $E[X|A] \rightarrow$ Conditional Expected Value of X given that event A has occured.

* For our question, this considers event where A is true i.e $-4 \leq x \leq 4$

$$E[X|A] = \int_a^b x \cdot f(x|A)\, dx.$$

$f(x|A)$ = Conditional probability of PDF $f_X(x)$, consider P(A) has occured.

$$= f_X(x) / P(A).$$

$$= 1/15 \div 8/15$$

$$= 1/8$$

$$E[X|A] = \int_a^b 1/8 \cdot x\, dx$$

$$= 1/8 \cdot \left[x^2/2\right]_{-4}^{4}$$

$$= 1/8 \cdot \left(\frac{4^2}{2} - \frac{-4^2}{2}\right)$$

$$= 1/8 \cdot 0$$

$$= 0 //$$

or.

only Considering where A has happened.

$$E[X|A] = \frac{a+b}{2} = \frac{-4+4}{2} = 0.$$

(b) $Var[X|A] \rightarrow$ Conditional variance of X given that event A has occured.

$$Var[X|A] = E[X^2|A] - (E[X|A])^2$$

As calculated earlier $E[X|A] = 0$

$$E[X^2|A] = \int_{-4}^{4} x^2 f_X(x)\,dx.$$

Recall $f_X(x) = \frac{1}{8}$

$$= \frac{1}{8} \cdot \left[\frac{x^3}{3}\right]_{-4}^{4}$$

$$= \frac{1}{8} \cdot \left[\frac{4^3}{3} - \frac{-4^3}{3}\right]$$

$$= \frac{1}{8} \cdot \frac{64+64}{3}.$$

$$= \frac{128}{8 \times 3}$$

$$= 16/3.$$

$$Var[X|A] = \frac{16}{3} - 0^2$$

$$= 16/3$$

or

$$Var[X|A] = \frac{(b-a)^2}{12} = \frac{(4-(-4))^2}{12}.$$

$$= \frac{\cancel{64}\ 16}{\cancel{12}\ 3}$$

$$= \frac{16}{3}$$

③ Exponential Random Variable Z with parameter $\lambda = 0.1$. Compute the probability $P(A|B)$, where $A = \{Z \geq 5 \text{ minutes}\}$ and $B = \{Z \geq 2 \text{ minutes}\}$.

Soln

For exponential random variable $f_Z(z) = \begin{cases} \lambda e^{-\lambda z} & z > 0 \\ 0 & \text{otherwise.} \end{cases}$

Applying Memoryless property. $P(Z > s+t \mid Z > s) = P(Z > t) = e^{-\lambda t}$

i.e if you've already waited s time units, the probability of waiting an additional t units is the same as if you started afresh.

Applying this

$$P(A|B) = P(Z \geq 2+3 \mid Z \geq 2) = P(Z \geq 3)$$

* Since she waited for $Z \geq 2$ first, she will wait for another 3 minutes to get to $Z \geq 5$.

$$P(A|B) = P(Z \geq 3)$$

$\lambda = 0.1$

$z = 3.$

$$P(A|B) = P(Z \geq 3) = e^{-\lambda z}$$

$$= e^{-(0.1)(3)}$$

$$= e^{-0.3}$$

$$= 0.0183$$

4). Let X represent the number of defective products returned on a given day,
- On Average 4 defective product are returned daily.
- The number of defective products returned follows a poisson distribution with $\lambda = 4$.

(a) Define the PMF of X

(b) Find $E[X]$ and $Var[X]$. Explain the significance of these values for warehouse operations.

Soln

a) PMF $= \begin{cases} \dfrac{\lambda^x e^{-\lambda}}{x!} & x = 0,1,2 \dots \\ 0 & \text{otherwise.} \end{cases}$

"for a poisson distribution, λ = mean, so $\lambda = 4$ represents that daily average of 4 defective returns".

$\lambda = 4$

$P(X=x) = P_X(x) = \begin{cases} \dfrac{4^x e^{-4}}{x!} & x = 0,1,2 \dots \infty \\ 0 & \text{otherwise.} \end{cases}$; $P_X(x) = \dfrac{4^x e^{-4}}{x!}$, $x = 0,1,2, \dots \infty$

(b) For a poisson distribution $E[X] = \lambda$

$E[X] = \sum_{x \in \Omega_x} x \cdot P_X(x)$ $\quad x \in \Omega_x$ = all values of x.

$E[X] = \sum_{x=1}^{\infty} x \cdot \dfrac{\lambda^x e^{-\lambda}}{x!}$ (Not starting with zero because it will give zero making it insignificant).

$= \sum_{x=1}^{\infty} \dfrac{x \cdot \lambda^x e^{-\lambda}}{x(x-1)!}$

$= \sum_{x=1}^{\infty} \dfrac{\lambda^x e^{-\lambda}}{(x-1)!}$

making $y = x-1$, $x = y+1$ for a distribution of y.

$E[X] = \sum_{y=0}^{\infty} \dfrac{\lambda^{y+1} e^{-\lambda}}{y!}$

$= \lambda \cdot \sum_{y=0}^{\infty} \dfrac{\lambda^y e^{-\lambda}}{y!}$; $\sum_{y=0}^{\infty} \dfrac{\lambda^y e^{-\lambda}}{y!} = 1.$

$= \lambda * 1$

$E[X] = \lambda$

here $E[X] = 4$

$Var[X] = \lambda$

Proving

$Var[X] = E[X^2] - (E[X])^2$

$E[X^2] = E[x(x-1)] + E[x]$

$Var[X] = E[x(x-1)] + E[X] - (E[X])^2$

$$E[x(x-1)] = \sum_{x \in \Omega_x} x(x-1) \cdot P_X(x)$$

$$= \sum_{x=2}^{\infty} x(x-1) \cdot \frac{\lambda^x e^{-\lambda}}{x!} \quad \text{starting at } x = 2.$$

$$= \sum_{x=2}^{\infty} \cancel{x(x-1)} \cdot \frac{\lambda^x e^{-\lambda}}{\cancel{x(x-1)}(x-2)!}$$

$$= \sum_{x=2}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-2)!} \quad \text{Let } y = x-2,\ x = y+2.$$

$$= \sum_{y=0}^{\infty} \frac{\lambda^{y+2} e^{-\lambda}}{y!}$$

$$= \lambda^2 \sum_{y=0}^{\infty} \frac{\lambda^y e^{-\lambda}}{y!}$$

$$= \lambda^2 \cdot \sum_{y=0}^{\infty} P_Y(y) \quad ; \sum_{y=0}^{\infty} P_Y(y) = 1$$

$$E[x(x-1)] = \lambda^2.$$

$$Var[X] = E[x(x-1)] + E[x] - (E[X])^2$$

$$= \lambda^2 + \lambda - \lambda^2$$

$$= \lambda$$

$\therefore Var[X] = \lambda = 4,$

The significance of Expected Value $E[X] = 4$ is that the management should ~~ma~~ make arrangements for an expectation of atleast 4 returned defects per day. ~~Arrangements in terms of 8~~ Here the management are to expect ~~an average~~ of 4 returns daily

It gives the management an insight of the average number of ~~defective~~ returns to expect daily.

Variance, $Var[X]$ which ~~will~~ is use to get the Standard deviation helps give insight on the spreadibility, how values could spread out from the expected value. It checks out the unpredictability or volatility of the process. In our case since the Variance is 4 Our Standard deviation is $\sqrt{Var} = \sqrt{4} = 2$. There would be a ~~predictability~~ spread out of $4-2, 4, 4+2 = 2, 4, 6$ i.e an expectation of 2-6 defective returns daily

(5) Follows a Normal distribution with mean $\mu = 500$ grams and Standard deviation $\sigma = 10$ grams

(a) Write the PDF of the normal distribution for this product weight. Identify $E[X]$ and $Var[X]$

(b) Calculate $P(490 \le x \le 510)$

(c) If the factory wants to ensure that 95% of the product weigh between 495 and 505 grams, what should σ be? What does this imply about improving the process?

Soln:

Normal distribution refers to Gaussian Distribution

- ~~Let X be a Gaussian (μ,~~

- Let X be a Guassian(μ, σ) Random Variable for the product weight.

then X is a Guassian$(500, 10)$ Random Variable for the product weight.

$$\text{PDF of } X \quad f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu_x)^2/2\sigma^2}$$

$$= \frac{1}{\sigma\sqrt{2\pi}} e^{\left[\frac{-(x-\mu_x)^2}{2\sigma^2}\right]}$$

from our question, $\mu_x = 500 \qquad \sigma = 10 \quad \therefore \sigma^2 = 100$

$$\left[f_X(x) = \frac{1}{10\sqrt{2\pi}} e^{-\frac{(x-500)^2}{200}} \right]$$

$E[X] = \text{mean } \mu_x = 500 \text{grams} = 500$

$Var[X] = (\text{Standard Deviation})^2 = \sigma^2 = 10^2 = 100$

(b) for $P(a \le x \le b) = \int_a^b f(x)\,dx$

$$P(a \le x \le b) = \phi\left(\frac{b-\mu_x}{\sigma}\right) - \phi\left(\frac{a-\mu_x}{\sigma}\right) \Big\} \text{(Standardization} \rightarrow \text{Standard normal using z-scores).}$$

~~$P(490 \le x \le 5) =$~~

$$\underline{P(490 \le x \le 510)} \implies \phi\left(\frac{510-500}{10}\right) - \phi\left(\frac{490-500}{10}\right)$$

$$\text{for } b = \phi\left(\frac{510-500}{10}\right)$$

$$= \phi(1)$$

$$\text{for } a = \phi\left(\frac{490-500}{10}\right)$$

$$= \phi(-1)$$

$$P(490 \le x \le 510) = \phi(1) - (\phi(-1))$$
$$= \phi(1) - (1 - \phi(1))$$
$$= \phi(1) - 1 + \phi(1)$$
$$= 2\phi(1) - 1 \longrightarrow \text{From the table } \phi(1) = 0.8413.$$
$$= 2(0.8413) - 1 = 1.6826 - 1 = 0.6826_{//}$$

or

using the emperical rule (68-95-99.7 Rule)

$$P(490 \le x \le 510) = P(-1 \le x \le 1)$$

68% falls within 1 standard deviation of the mean.

$$\therefore P(490 \le x \le 510) = 68\%$$
$$= 0.68_{//}$$

(c) For 95%, $P(-2 \le x \le 2)$ i.e 2 standard deviation from the mean

i.e $P(\mu - 2\sigma \le x \le \mu + 2\sigma)$.

If products weigh between 495 and 505

$$P(495 \le x \le 505) = P(\mu - 2\sigma \le x \le \mu + 2\sigma) \quad \text{for } 95\%$$

$495 = \mu - 2\sigma$	$\mu = 500.$	$505 = \mu + 2\sigma$
$495 = 500 - 2\sigma$		$505 = 500 + 2\sigma$
$495 - 500 = -2\sigma$		$505 - 500 = 2\sigma$
$\cancel{-}5 = \cancel{-}2\sigma$		$5 = 2\sigma$
$\sigma = 2.5$		$\sigma = 2.5$

→ This implies that in production, atleast 95% of the products should be in the range of 495 to 505 grams, that is lice ±5 grams ~~against the current~~ difference with the mean of 500, against 68% being ~~490 to 510 grams~~ between 490 to 510 grams. This calls for more improvement in the process, as precision and tighter control of the process will be required to get an improved result.

⑥ A machine part fails according to an exponential distribution with mean time between failures of 10 hours. Let X represent the time (in hours) until the next failure

(a) Write the PDF for the exponential distribution.

(b) Derive $E[X]$ & $Var[X]$

Soln.

(a) X represents the time (in hours) until the next failure [time between a poisson Process.

PDF of an exponential random variable X with rate λ (where $\lambda > 0$).

$$f(x;\lambda) = \lambda e^{-\lambda x} \quad ; x > 0$$

* Mean time between failure = 10 hours.

X = time (in hours) until the next failure

$\lambda = \frac{1}{10}$

$= 0.1$ failures per hour.

$$\underline{f(x;\lambda) = 0.1 e^{-0.1x} \quad ; x > 0}$$

(b) $E[X] = \int_0^\infty x f(x)\, dx$

$= \int_0^\infty \lambda e^{-\lambda x} x \, dx$

$= \lambda \int_0^\infty x \cdot e^{-\lambda x} \, dx$

let $u = x$ $\quad du = dx \quad dv = e^{-\lambda x} dx \quad v = -\frac{1}{\lambda} e^{-\lambda x}$

Recall that $\int u\,dv = uv - \int v\,du$

$$\int_0^\infty x \cdot e^{-\lambda x} dx = \left. -\frac{x}{\lambda} \cdot e^{-\lambda x} \right]_0^\infty - \int_0^\infty -\frac{1}{\lambda} e^{-\lambda x} \cdot dx$$

$$\int_0^\infty x \cdot e^{-\lambda x} dx = \left(-\frac{\infty}{\lambda} \cdot e^{-\lambda(\infty)} - -\frac{0}{\lambda} \cdot e^{-\lambda(0)} \right) + \frac{1}{\lambda} \int_0^\infty e^{-\lambda x} dx$$

$$= 0 + \frac{1}{\lambda} \int_0^\infty e^{-\lambda x} dx$$

$$= \frac{1}{\lambda}\int_0^{\infty} e^{-\lambda x}\,dx$$

$$= \frac{1}{\lambda}\left[-\frac{1}{\lambda}e^{-\lambda x}\right]_0^{\infty}$$

$$= \frac{1}{\lambda}\left[-\frac{1}{\lambda}e^{-\lambda(\infty)} - \left(-\frac{1}{\lambda}e^{-\lambda(0)}\right)\right].$$

$$= \cancel{\frac{1}{\lambda}\left(\frac{1}{\lambda}\right)}.\ \frac{1}{\lambda}\left[0 + \frac{1}{\lambda}\right]$$

$$\int_0^{\infty} x \cdot e^{-\lambda x}\,dx = \frac{1}{\lambda^2}$$

$$E[X] = \lambda\int_0^{\infty} x \cdot e^{-\lambda x}\,dx$$

$$= \lambda\left(\frac{1}{\lambda^2}\right)$$

$$= \frac{1}{\lambda}$$

~~E[X]~~ Recall $\lambda = 0.1$

$$E[X] = \frac{1}{0.1}$$

$$= 10_{//}$$

$$\underline{Var[X] = E[x^2] - [E[X]]^2.}$$

$$E[X^2] = \int_0^{\infty} x^2 \cdot f(x)\,dx.$$

$$= \int_0^{\infty} x^2 \cdot \lambda e^{-\lambda x}\,dx$$

$$= \lambda\int_0^{\infty} x^2 \cdot e^{-\lambda x}\,dx$$

$$\int_0^{\infty} x^2 \cdot e^{-\lambda x}\,dx = uv - \int_0^{\infty} v\,du$$

$$u = x^2 \quad du = 2x\,dx \qquad dv = e^{-\lambda x}\,dx \quad v = -\frac{1}{\lambda}e^{-\lambda x}$$

$$\int_0^{\infty} x^2 \cdot e^{-\lambda x}\,dx = x^2 \cdot -\frac{1}{\lambda}e^{-\lambda x}\Big]_0^{\infty} - \int_0^{\infty} -\frac{1}{\lambda}e^{-\lambda x} \cdot 2x\,dx.$$

$$= -\frac{x^2}{\lambda}e^{-\lambda x}\Big]_0^{\infty} + \frac{2}{\lambda}\int_0^{\infty} e^{-\lambda x} \cdot x\,dx.$$

$$= \left(\frac{-\infty^2}{\lambda} e^{-\lambda(\infty)} - -\frac{0^2}{\lambda} e^{-\lambda(0)} \right) + \frac{2}{\lambda} \int_0^\infty x \cdot e^{-\lambda x} dx$$

$$= 0 + \frac{2}{\lambda} \int_0^\infty x \cdot e^{-\lambda x} dx$$

$$\int_0^\infty x^2 \cdot e^{-\lambda x} dx = \frac{2}{\lambda} \int_0^\infty x \cdot e^{-\lambda x} dx$$

recall $\int_0^\infty x \cdot e^{-\lambda x} dx = \frac{1}{\lambda^2}$

$$\int_0^\infty x^2 \cdot e^{-\lambda x} dx = \frac{2}{\lambda} \cdot \frac{1}{\lambda^2}$$

$$= \frac{2}{\lambda^3}$$

$$E[x^2] = \lambda \int_0^\infty x^2 \cdot e^{-\lambda x} dx$$

$$= \cancel{\lambda} \cdot \frac{2}{\cancel{\lambda^3}\lambda^2}$$

$$= \frac{2}{\lambda^2}$$

$$Var[X] = E[x^2] - (E[X])^2$$

$$= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2$$

$$= \frac{2}{\lambda^2} - \frac{1}{\lambda^2}$$

$$= \frac{2-1}{\lambda^2}$$

$$= \frac{1}{\lambda^2}.$$

Recall $\lambda = 0.1$

$$Var[X] = \frac{1}{(0.1)^2}$$

$$= \frac{1}{0.01}$$

$$= 100 //$$